

Stop and Wait Protocol: Performance and Problem-Solving Analysis

Executive Summary

The Stop and Wait (S&W) protocol is a basic flow control mechanism in which the sender transmits one frame and waits for an acknowledgment (ACK) before sending the next. Its performance is fundamentally governed by the relationship between:

- **Transmission Delay (T_{trans})**
- **Propagation Delay (T_{prop})**

Key insights:

- **Efficiency Formula:** The most accurate efficiency expression is $\eta = \text{Useful Time} / \text{Total Time}$. Simplified formulas such as $1 / (1 + 2a)$ are valid only when additional delays are negligible.
- **Impact of Link Distance:** Larger propagation delay (longer distance) significantly reduces efficiency.
- **Impact of Bandwidth:** Increasing bandwidth reduces transmission delay, which can decrease utilization if propagation delay remains unchanged.
- **Wireless Reliability:** With packet error probability p , the average number of transmissions per packet is $1 / (1 - p)$.

1. Fundamental Principles and Mathematical Framework

1.1 Delay Components

For one successful frame transmission, total time consists of:

Transmission Delay (T_{trans})

Time to push all frame bits onto the link.

$$T_{\text{trans}} = \text{Frame Size} / \text{Bandwidth}$$

Propagation Delay (T_{prop})

Time for a bit to travel from sender to receiver.

$$T_{\text{prop}} = \text{Distance} / \text{Propagation Velocity}$$

Processing Delay (T_{proc})

Time taken by receiver to process the frame.

Queuing Delay (T_{que})

Time spent in buffers or queues.

ACK Transmission Delay (T_{ACK})

Time to transmit the acknowledgment.

$$T_{ACK} = \text{ACK Size} / \text{Bandwidth}$$

1.2 Efficiency Equation

Efficiency (η) is defined as:

$$\eta = \text{Useful Time} / \text{Total Time}$$

Where:

$$\text{Useful Time} = T_{frame}$$

Total Time =

$$T_{frame} + 2T_{prop} + T_{que} + T_{proc} + T_{ACK}$$

If queuing delay, processing delay, and ACK transmission delay are negligible:

$$\eta = T_{frame} / (T_{frame} + 2T_{prop})$$

1.3 Throughput

Throughput represents the actual rate of successful data delivery:

$$\text{Throughput} = \eta \times \text{Bandwidth}$$

OR

$$\text{Throughput} = \text{Frame Size} / \text{Total Time}$$

2. Link Utilization and Performance Analysis**2.1 Conditions for Low Link Utilization**

Link utilization is lowest when:

Long Link Length

Increasing distance increases propagation delay (T_{prop}), enlarging the denominator in the efficiency equation.

High Transmission Rate (Bandwidth)

Higher bandwidth reduces T_{trans} . Since T_{trans} is the useful time, reducing it while T_{prop} remains constant lowers efficiency.

2.2 Condition for 50% Efficiency

For $\eta \geq 50\%$:

$$T_{trans} \geq 2T_{prop}$$

This ensures the transmission time is at least equal to the round-trip propagation delay.

3. Case Studies and Numerical Problem Solving

3.1 Equal Data and ACK Size Scenario

Given:

$$T_{trans_frame} = T_{trans_ACK} = a$$

$$\text{Propagation delay} = b$$

Total cycle time:

$$= a (\text{Data } T_{trans}) + b (\text{Data } T_{prop}) + a (\text{ACK } T_{trans}) + b (\text{ACK } T_{prop})$$

$$= 2a + 2b$$

The second packet can be sent after $2a + 2b$ time.

3.2 Throughput Calculation Example

Given:

- Frame size = 1000 bytes = 8000 bits
- Sender bandwidth = 80 Kbps

- ACK size = 100 bytes = 800 bits
- Receiver bandwidth = 8 Kbps
- Propagation delay = 100 ms

Step 1: Transmission delay of frame

$$T_{\square_frame} = 8000 / 80000 = 0.1 \text{ s}$$

Step 2: Transmission delay of ACK

$$T_{\square_ACK} = 800 / 8000 = 0.1 \text{ s}$$

Step 3: Total time

$$= 0.1 + (2 \times 0.1) + 0.1$$

$$= 0.4 \text{ s}$$

Step 4: Throughput

$$= 8000 / 0.4$$

$$= 20000 \text{ bps}$$

Final result:

$$= 2500 \text{ bytes/second}$$

3.3 Efficiency with Overheads (GATE 2017 Type)

Given:

Payload = 1980 bytes

Header = 20 bytes

Total frame size = 2000 bytes = 16000 bits

Bandwidth = 1 Mbps

Transmission delay:

$$T_{\square} = 16000 / 1,000,000 = 16 \text{ ms}$$

Total time includes:

- $T_{\square} = 16 \text{ ms}$
- $2T_{\square} = 1.5 \text{ ms}$
- Processing delay = 0.25 ms
- ACK transmission delay = 0.16 ms

Total time = 17.91 ms

Efficiency:

$$\eta = 16 / 17.91$$
$$\approx 89.33\%$$

4. Reliability and Error Probability

4.1 Average Number of Transmission Attempts

If packet error probability is p :

$$\text{Average transmissions per packet} = 1 / (1 - p)$$

For n packets:

$$\text{Average total transmissions} = n / (1 - p)$$

Examples:

If $n = 100$ and $p = 0.2$:

$$= 100 / 0.8$$

$$= 125 \text{ transmissions}$$

If average transmissions = 500 and $p = 0.6$:

$$\text{Unique packets sent} = 500 \times (1 - 0.6)$$

$$= 200 \text{ packets}$$

5. Strategic Recommendations for Problem Solving

Avoid Oversimplified Formulas

Use the complete efficiency formula unless delays are explicitly negligible.

Maintain Unit Consistency

Convert all quantities to consistent units before calculation.

Check Throughput Units Carefully

Confirm whether the answer is required in bits per second (bps) or bytes per second.

Frame Size for Target Utilization

For a target efficiency (e.g., 75%):

1. Set up inequality:
 $\eta \geq 0.75$
2. Substitute full efficiency formula.
3. Solve for T_{frame} .
4. Compute required frame size using:
Frame Size = $T_{\text{frame}} \times \text{Bandwidth}$

